# ADGSTUDIOS

## DisputeGuard

**Engine**

### Test Report

Automated verification of the equations and flows

**PASS** 50 / 50 checks passed

Generated October 2026 — Rust reference engine

*Companion to the DisputeGuard Equations and Algorithms guide.*

## Summary

| Metric | Value |
|---|---|
| Total checks | 50 |
| Passed | 50 |
| Failed | 0 |
| Sections | 11 |

This report is produced by the `report` binary in the `disputeguard-engine` crate. Each row runs a deterministic check against the reference implementation and records the observed value.

## Money and Exact Amounts 5/5

Amounts are whole minor units; precision follows the currency and floating-point rounding never appears.

| Check | Observed | Result |
|---|---|---|
| R1234.56 parses to cents | `minor = 123456` | **PASS** |
| 0.10 + 0.20 is exact | `minor = 30 (= 0.30)` | **PASS** |
| JPY rejects fractional input | `parse("1.5", JPY) = BadPrecision` | **PASS** |
| BHD keeps three decimals | `minor = 1234` | **PASS** |
| Cross-currency add is refused | `ZAR + USD = CurrencyMismatch` | **PASS** |

## Ownership and Permission 4/4

View depends on ownership or an analyst/admin role; dispute creation always requires ownership.

| Check | Observed | Result |
|---|---|---|
| Owner can view | `CanView(owner, t) = true` | **PASS** |
| Stranger cannot view | `CanView(stranger, t) = false` | **PASS** |
| Analyst can view | `role grants CanView` | **PASS** |
| Only owner can create | `analyst cannot create a dispute` | **PASS** |

## Dispute Lifecycle and Concurrency 4/4

Only authorised transitions proceed; stale versions are rejected and successful changes bump the version.

| Check | Observed | Result |
|---|---|---|
| SUBMITTED + BEGIN_REVIEW -> UNDER_REVIEW | `authorised transition` | **PASS** |
| SUBMITTED + ISSUE_REFUND -> bottom | `refund is never a lifecycle transition` | **PASS** |
| Version bumps on success | `v 3 -> 4` | **PASS** |

| Check | Observed | Result |
|---|---|---|
| Stale version is rejected | `expected 2 != actual 3` | **PASS** |

## Dispute Creation: Idempotency and Uniqueness 4/4

Retries return the original result; at most one active dispute may exist per transaction.

| Check | Observed | Result |
|---|---|---|
| Retry replays original dispute | `ids 1 == 1, replayed = true` | **PASS** |
| Key reuse with new body refused | `KeyReuseWithDifferentRequest` | **PASS** |
| Second active dispute refused | `ActiveDisputeExists` | **PASS** |
| Active count is 1 | `count = 1` | **PASS** |

## Deterministic Risk Indicators 12/12

Each signal has its own safe guards; a missing input is reported as unavailable rather than treated as low risk.

| Check | Observed | Result |
|---|---|---|
| Amount signal bounded in [0,1] | `A(9000; x_max=10000) = 0.9886` | **PASS** |
| Amount signal clamps at scale | `A(1e6; x_max=10000) = 1.0000` | **PASS** |
| z-score flags R900 outlier | `z = 46.378 (> 3)` | **PASS** |
| z-score unavailable at zero variance | `constant history -> None` | **PASS** |
| Robust z (MAD) flags outlier | `z* = 45.023` | **PASS** |
| MAD = 0 is undefined (no div-by-zero) | `constant history -> None` | **PASS** |
| Velocity needs positive window | `V(12, 0) = None` | **PASS** |
| Haversine JHB->CPT ˜ 1262 km | `d = 1261.6 km` | **PASS** |
| Impossible travel velocity detected | `v_required = 1262 km/h over 1 h` | **PASS** |
| Combined score in [0,1] | `R = 0.440` | **PASS** |
| Missing signal refuses a score | `geography = None -> MissingSignal` | **PASS** |
| Weights must sum to 1 | `sum = 1.3 -> InvalidWeights` | **PASS** |

## Signal Normalisation and Integration 5/5

Raw anomaly measures map into bounded [0,1] signals, and the full calculation returns a score or an explicit reason.

| Check | Observed | Result |
|---|---|---|
| History outlier saturates signal | `history_signal(R900) = 1.000` | **PASS** |
| Velocity z-score flags burst | `z_V(2, 0.5, 5) = 6.0` | **PASS** |
| Merchant signal is chargeback ratio | `merchant_signal(3, 12) = 0.25` | **PASS** |
| calculate_risk scores full inputs | `score = 0.238` | **PASS** |

| Check | Observed | Result |
|---|---|---|
| Missing input gives reason not zero | `score = unavailable (merchant)` | **PASS** |

## Role Gating and Audit Log 4/4

Only authorised roles drive the case, and every committed change is recorded in an ordered audit log.

| Check | Observed | Result |
|---|---|---|
| Customer cannot begin review | `actor_may_perform(Customer, BeginReview) = false` | **PASS** |
| Analyst can begin review | `actor_may_perform(Analyst, BeginReview) = true` | **PASS** |
| Audit log records created + 3 transitions | `4 ordered events` | **PASS** |
| Customer review attempt refused | `NotAuthorised, no event added` | **PASS** |

## Evidence Upload Workflow 4/4

A file is validated for authorisation, size, declared and detected type, and scan status before it is recorded.

| Check | Observed | Result |
|---|---|---|
| Clean PDF accepted with server name | `storage_name = srv-abc123` | **PASS** |
| Detected type overrides claim | `DetectedTypeNotAllowed` | **PASS** |
| Scanner unavailable != clean | `ScannerUnavailable` | **PASS** |
| Unauthorised upload refused first | `NotAuthorised` | **PASS** |

## Safe Background Retries 3/3

A redelivered job replays its durable result exactly once; permanent failures are not retried.

| Check | Observed | Result |
|---|---|---|
| Duplicate delivery replays original | `Completed then Replayed(refund-ok)` | **PASS** |
| Non-retryable job refused | `NotRetryable` | **PASS** |
| Permanent failure is not retried | `FailedForReview` | **PASS** |

## Bounded AI Investigation 3/3

The loop gathers only allowlisted, read-only observations and stops on budget or a disallowed tool, leaving the case unchanged.

| Check | Observed | Result |
|---|---|---|
| Complete run reaches human review | `stage = Review, tool_calls = 3` | **PASS** |
| Budget stops before 3rd tool call | `stage = Stopped, tool_calls = 2` | **PASS** |
| Disallowed tool blocked pre-execution | `tool_calls = 1, observations = 1` | **PASS** |

## Sample Transaction Data 2/2

The demo transactions shared with the web landing page use unique references and exact positive minor units.

| Check | Observed | Result |
|---|---|---|
| References are unique | `9 transactions, 9 unique ids` | **PASS** |
| All amounts positive minor units | `every amount_minor > 0` | **PASS** |

## Appendix: Sample Transactions

The nine demo transactions used as fixtures (amounts shown as exact ZAR from integer cents).

| Ref | Merchant | Category | Amount (ZAR) |
|---|---|---|---|
| TX-0042 | Atlas Online | Shopping | 1249.00 |
| TX-0043 | Corner Coffee | Food & drink | 68.00 |
| TX-0044 | Fresh Market | Groceries | 432.50 |
| TX-0051 | Studio Supply | Shopping | 899.00 |
| TX-0052 | Stream House | Subscription | 149.00 |
| TX-0053 | Cloud Desk | Software | 229.00 |
| TX-0061 | Coastline Stay | Accommodation | 2450.00 |
| TX-0062 | City Shuttle | Transport | 185.00 |
| TX-0063 | Platform Café | Food & drink | 96.00 |

*All figures are illustrative. Risk indicators are advisory and never decide a dispute.*